

Francisco Medina

Prof. Sridhar Ganti

ITAI 2372: AI Applications

13 June 2026

Smart Cities: AI in Healthcare and Agriculture

1. Industry Overview

Healthcare and agriculture might not seem like they belong in the same conversation, but when you zoom out and look at what makes a society actually function, they are probably the two most foundational industries that exist. Healthcare keeps people alive and healthy, and agriculture keeps them fed, and right now both are under enormous pressure.

On the healthcare side, the global demand for medical care is outpacing the supply of trained specialists. Conditions like diabetes, which affects over 500 million people worldwide, create downstream complications like diabetic retinopathy that require specialized eye care to catch early before they cause permanent blindness. The problem is that ophthalmologists require 8 to 12 years of specialized training and are already scarce, especially in rural and low-income areas where diabetes rates tend to be highest. In agriculture, the challenge is feeding a global population that keeps growing while climate change makes weather patterns less predictable and critical water sources like the Ogallala Aquifer beneath the Texas Great Plains are being depleted from overuse. Farmers are being asked to produce more food from the same land with fewer resources and far less predictability than the previous generation dealt with. Both industries are

at a point where traditional approaches are no longer sufficient, which is exactly why AI has become so relevant.

2. Current AI Trends

Two of the most active areas where AI is making a concrete difference right now are medical diagnostics and precision agriculture, and both have real, deployable examples already operating in the world.

In healthcare, one of the clearest applications is AI-powered retinal disease diagnosis. Google DeepMind developed an AI system that analyzes photographs of the retina to detect diabetic retinopathy, a leading AI-preventable cause of blindness. The system was trained on 128,000 labeled retinal photographs reviewed by 54 ophthalmologists, and it achieved 90% sensitivity and 98% specificity in trials, meaning it correctly identifies 90 out of 100 people who actually have the disease and correctly clears 98 out of 100 healthy patients. In 2019 and 2020, this AI was deployed in real rural clinics in Thailand where no ophthalmologist was present, screening thousands of diabetic patients who would otherwise have had no access to early detection. The FDA approved the first fully autonomous AI diagnostic system, IDx-DR, back in 2018, which was a landmark moment because it was the first system that could return a diagnosis without a doctor reviewing it. More recently, researchers at AIM Doctor right here in Houston developed a tool called SMART (Simple Mobile AI Retina Tracker) that runs on a basic smartphone and detects diabetic retinopathy with over 99% accuracy in under one second, trained on diverse retinal images from six continents. AI is also being used across medicine to

detect early signs of diabetes in blood samples and to scan large volumes of imaging data for cancer, processing thousands of scans far faster than any radiologist could.

In agriculture, AI weather forecasting has become one of the most practical and high-value tools available to farmers. Companies like The Climate Corporation, which operates a product called FieldView, and IBM through its Weather Company division now offer hyperlocal AI weather forecasting built specifically for individual farms. These systems pull from soil moisture sensors on the ground, satellite imagery tracking cloud cover and crop health, and 30 to 50 years of historical weather records for a specific location to generate predictions hour by hour up to 14 days out. For a wheat farmer in Lubbock, Texas, that level of precision means knowing whether a late spring frost is coming to their specific 400 acres rather than just knowing it might get cold somewhere in the region. Research shows that AI-guided precision irrigation typically reduces water use by 30 to 50% compared to traditional schedule-based irrigation, which matters enormously for a region drawing from a depleting aquifer. Drones are also being used to map farm terrain and identify the exact areas where pesticides or nutrients need to be applied, cutting chemical use and reducing runoff.

3. Ethical Implications

As useful as these tools are, they come with serious ethical complications worth taking seriously.

The most significant concern in both industries is bias rooted in unrepresentative training data. AI systems learn from whatever data they were given, and if that data does not reflect the

full diversity of people who will eventually use the system, the results are unequal by design. Google's retina AI was trained on 128,000 images, but the question of how demographically diverse those images were is legitimate and important. AI diagnostic tools used in dermatology and cardiology are already documented to perform worse on patients of color, and retinas can vary in pigmentation across ethnic groups. When a system trained mostly on data from wealthy countries gets deployed in Thailand, sub-Saharan Africa, or rural communities in the American South, there is a real risk that it performs worse for the people who need it most, meaning a false all-clear from a system that was never properly validated on their population. In agriculture, the same problem plays out differently but with similar consequences. Weather forecasting models trained on well-studied regions in North America and Europe may not generalize as well to smallholder farms in East Africa or South Asia where microclimates and agricultural patterns differ significantly.

The second major concern is privacy and data ownership, particularly in healthcare. When a rural patient in Thailand or Houston gets their retina photographed and that image is sent to a cloud system operated by a company like Google, there are real questions about who actually owns that data. Medical records can be used by insurance companies to deny coverage or raise premiums, and in countries without strong privacy protections, that kind of health data could be used to discriminate against people based on medical status. Patients in rural clinics often lack the information needed to make genuinely informed decisions about what they are consenting to, especially when the system analyzing their data is operated by a foreign tech company with commercial interests. A related issue is the digital divide. An AI weather service that costs \$1,500 per year looks completely different depending on who is paying for it. For a corporate farm operating 50,000 acres, that cost is negligible. For a smallholder farmer in rural

Kenya farming two acres with a total annual income of a few hundred dollars, it is completely out of reach. If access to these tools is determined entirely by ability to pay, AI ends up widening the competitive gap between large industrial agriculture and small-scale farming rather than helping address global food insecurity.

4. Future Trends

Looking 5 to 10 years ahead, both healthcare and agriculture are likely to be transformed in ways that go well beyond what is currently deployed.

In healthcare, the trajectory points toward AI that does not just detect existing disease but predicts it before symptoms appear. The combination of AI with genomics and gene therapy opens up the possibility of systems that analyze a person's genetic profile alongside their medical history and flag elevated risks years before a condition manifests. Diagnostic AI will likely expand far beyond the retina into broader disease detection across cardiology, oncology, and metabolic conditions. As these systems become more accurate and widely used, the bigger challenge will be governance: who regulates them, how do you ensure they work equitably across diverse populations, and how much autonomy do you give an AI in a medical decision. The FDA approving IDx-DR as an autonomous system in 2018 was the beginning of that conversation, not the end of it.

In agriculture, the next decade will likely bring fully or semi-autonomous farming systems where AI does not just advise the farmer but directly controls equipment in real time. If a system predicts rain in six hours, it could automatically pause irrigation; if it detects a late frost

building, it could trigger protective equipment without waiting for a human to check an app. Satellite and drone technology will become even more integrated, allowing AI to monitor crop health at the individual plant level across thousands of acres simultaneously. Climate change will actually increase the value of these tools because the historical patterns that current models rely on are already shifting, and AI systems that update continuously from real-time data will adapt faster than any static model or human intuition could. The risk is that if access remains unequal, the gap between large commercial agriculture and small subsistence farming widens permanently, with serious consequences for global food security.

5. My Reflection

What surprised me most about both of these industries is how close the technology already is to having a massive real-world impact, and how much of the barrier to that impact is not technical but logistical and ethical. I came in thinking the main challenge was building AI that works well enough, but reading through the retinal AI research and the precision agriculture case studies, the systems are already remarkably accurate. Google's retina AI matches specialist doctors. The SMART tool from AIM Doctor here in Houston hits 99% accuracy on a smartphone. FieldView is already saving farmers real money on irrigation and crop protection. The harder problems are making sure these tools actually reach the people who need them, making sure they work equally well across different populations, and making sure the data they collect serves patients and farmers rather than just the companies deploying the systems.

Would I want to work in these industries? Yeah, honestly I would, specifically at the intersection where AI systems are built and where the real-world deployment challenges get

figured out. The retinal AI project or something like the SMART tool is exactly the kind of problem I find interesting, building a system that has to work in a resource-constrained environment for a population that does not have alternatives. Agriculture AI feels similar in that the constraints are real and the stakes are high. Those are the kinds of problems that make the work feel like it actually matters.